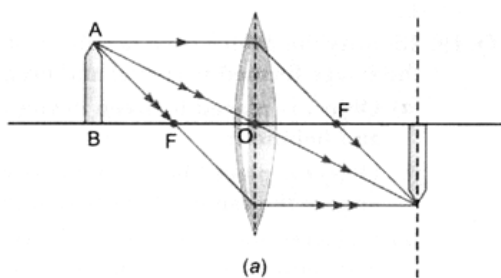


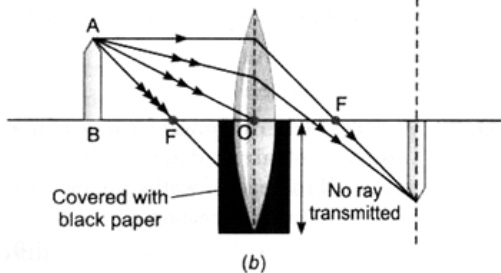
Solution

LIGHT -06

Class 10 - Science



1.



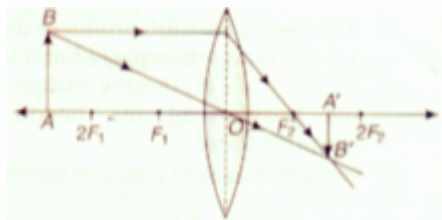
Yes, even when one-half of a convex lens is covered with a black paper, the lens will produce a complete image.

Take a live candle, keep it in front of a convex lens mounted on an optical bench.

Move the candle along the axis of bench and take its full image on a screen. Now cover the lower half of lens with a black paper without changing the positions of candle, lens and screen.

You will observe that full image of candle is still seen on the screen, but the intensity of image is reduced. The reason is that a large number of rays incident on the lens are blocked. In the case of covered lower half of lens with black paper, the rays that are emerging from candle and incident on lens are refracted from upper part only and form the full image.

2. i. The position of object AB would have been beyond $2F_1$.



ii. Size of the object would have been bigger than the size of image.

3. a. The point from which parallel rays of light, after refraction from a lens, appear to diverge is called focus of divergent lens, and the distance between optical centre and this focus of a divergent lens is called focal length of divergent lens.

b. Focal length of divergent lens, $f = -30$ cm

Image distance, $v = -15$ cm

Object height, $h_1 = 6$ cm

We know that, $\frac{1}{f} = \frac{1}{v} - \frac{1}{u}$

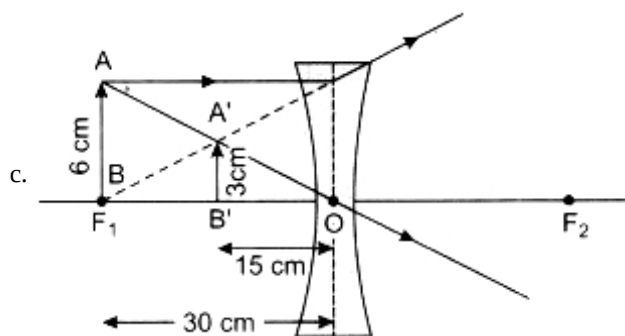
or $\frac{1}{u} = \frac{1}{v} - \frac{1}{f}$

$$\Rightarrow u = \frac{vf}{f-v} = \frac{-15 \times -30}{-30+15}$$

$$= \frac{450}{-15} = -30 \text{ cm}$$

$$\text{Now, } m = \frac{h_2}{h_1} = \frac{v}{u}$$

$$\Rightarrow h_2 = \frac{v}{u} \times h_1 = \frac{-15}{-30} \times 6 = 3 \text{ cm}$$

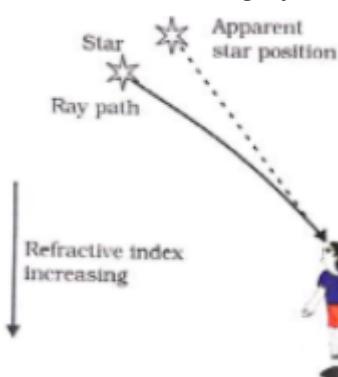


4. Let us assume that the window pane is between F_2 and infinity from this lens and this is a convex lens. We know that when the object is between infinity and F_2 , its inverted and real image is formed between $2F$ and $2F_2$.

Now, the distant building is at infinity from the lens. Its image would be formed at $2F$. So, the screen needs to be moved towards the lens in order to get a sharp image. Its approximate focal length is 10 cm (less than image distance in earlier case).

5. **Atmospheric refraction:-** The refraction of light caused by the earth's atmosphere (having their layers of varying optical densities) is called atmospheric refraction.

Light from a star is refracted as it leaves space and enters the earth's atmosphere. Air higher up in the sky is rare but that near the Earth's surface is denser. So, as the light from a star comes down the dense air bends the light more. Therefore, the apparent position to the star is slightly different from its actual position.



6. Refractive indices of Air, Ice and Benzene are 1.003, 1.31 and 1.5 respectively. Velocity of light in a medium is inversely proportional to refractive index of the medium. Light will travel fastest in air (having least refractive index i.e. 1.0003) and slowest in Benzene (having maximum refractive index i.e. 1.5)

7. (i) Focal length = $1/\text{power} = 1/5D = 1/5m = 20\text{cm}$

It is convex lens of focal length 20cm.

So, the Magnified image will be formed in all cases, 20cm is focus, 18 cm is on focal length, 22 cm and 30 cm is between focus and focus and center of curvature. In all cases, magnified image is formed.

(ii) In case of 22 cm and 30 cm image formed is real and hence can be obtained on screen.

8. i. a) Shaving mirror- Concave mirror

b) Rear view mirror - Convex mirror

c) Reflection in search-lights - Concave mirror.

ii. The three differences are:

a) Real image can be obtained on screen but a virtual image cannot be obtained.

b) Reflected/Refracted rays actually meet where the real image is formed while for virtual they only appear to meet.

c) A Real image is always inverted while the virtual image is always erect.

9. The virtual image formed by a concave mirror is always magnified whereas the virtual image formed by a convex mirror is diminished.

10. 1. If image is of same size, laterally inverted and erect, it is plane mirror.
2. If image is bigger or smaller in size and inverted or erect, it is concave mirror.
3. If image is smaller in size and erect as in rear view mirror, it is convex mirror.

11. i. a. Object should be between pole and focus.

b. At the focus.

ii. a. Small head-convex mirror.

b. Legs of normal size-plane mirror.

12. We know that pencil appears to be bent at the interface of air and water because of refraction of light. The degree of refraction depends on refractive index of a given liquid. Refraction indices of kerosene, water and other liquids would be different. Hence, degree of bend would be different in case of different liquids.

13.	Real Image	Virtual Image
	1. It is formed by the actual meeting of reflected (or refracted) ray.	1. It is formed when reflected (or refracted) rays appear to meet when produced backward
	2. It can be obtained on the screen.	2. It cannot be obtained on the screen.
	3. It is always inverted.	3. It is always erect.
	4. It is formed by concave mirror or convex lens.	4. It is formed by concave, convex and plane mirror (or concave and convex lens.)

14. When an object is placed between Focus and Pole of concave mirror, the image formed is virtual, magnified, erect and behind the mirror. When an object is placed between Curvature and Focus of concave mirror, the image formed is real, magnified, inverted at the same side of mirror.

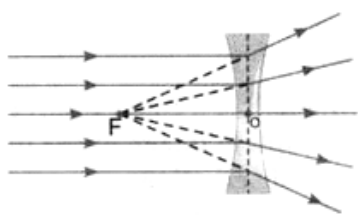
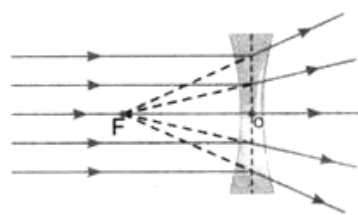
15. $P = \frac{1}{f}$, $P \propto \frac{1}{f}$

The power of the lens is inversely proportional to the focal length of the lens. A lens with the focal length 20 has more power than a lens with a focal length of 40 cm.

Therefore, a lens with higher power should be used to obtain more convergent light.

16. Given focal length f of lens 20 cm

To obtain real and magnified image, the object should be placed between F_1 and $2F_1$. So the range will be from 20 cm to 40 cm of convex lens.

17.	Convex Lens	Concave Lens
	1. It is thicker at the center than at the edges. 2. It has real Focus. 3. It converges a parallel beam of light on refraction through it.	1. It is thinner at the center than at the edges. 2. It has virtual focus. 3. It diverges a parallel beam of light on refraction through it.
	4. 	4. 

18. i. Concave Mirror

ii. It is given, object distance $u = -20\text{m}$, distance $v = 80\text{m}$

Magnification is given as $M = -v/u$

$$M = -v/u = -(-80\text{m}/-20) = 4$$

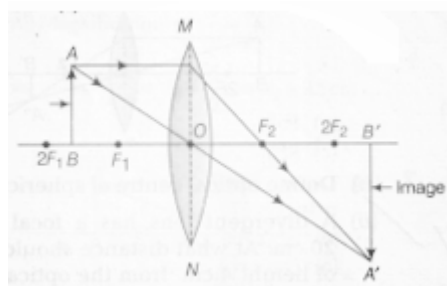
iii. Distance between object and image $v - u = -80\text{m} - (-20\text{m}) = -60\text{m}$

19. i. a. Object is placed between F and $2F$ of thin converging lens.

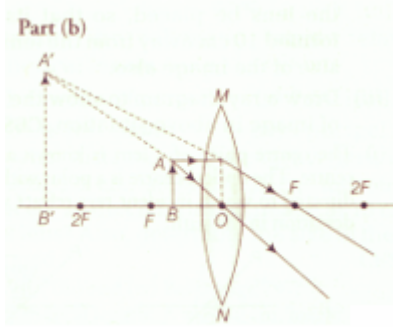
b. Object is placed between optical centre and F .

ii. The ray diagrams for real magnified and virtual magnified images are as follows:

Part (a)



Part (b)

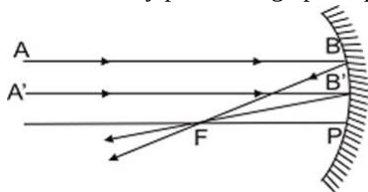


- iii. a. There will be no change in focal length of converging lens.
- b. Intensity will become one-fourth and brightness of lens will be less .

20. Images formed by concave mirror.

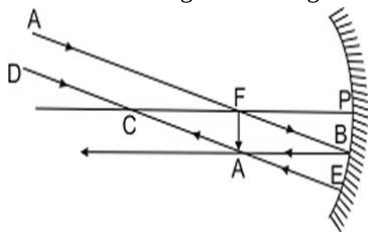
i. Object at Infinity. Two cases arise :

- a. When mirror is in parallel plane to the object. In such a case, rays from infinity come parallel to principal axis. After reflection they pass through principal focus F (Rule 1). Image is extremely small, it is real, inverted and at principal focus.



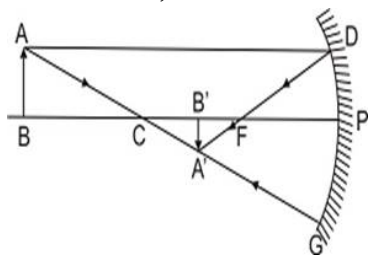
Object at infinity, real extremely diminished image is formed at principal focus.

- b. When mirror is inclined so that the rays strike the mirror obliquely. The ray AB passing through F after reflection goes parallel to principal axis towards BA' (Rule 2). Another ray DE through C striking the mirror at E is reflected back. The two form an image at A' Image is real, inverted, extremely diminished and at F.



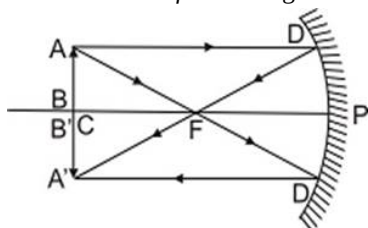
Object at infinity, image at F. It is real, inverted, very much diminished.

- c. Object beyond C. A ray AD from A parallel to principal axis after reflection passes through F (Rule 1), Another ray from A through C, ray AG is reflected back along the same path (Rule 3), forming real, diminished, inverted image of AB is formed at A'B', between F and C.



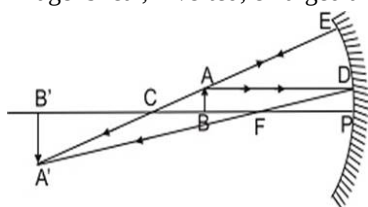
Object beyond C, a real, inverted diminished image between F and C, inverted at C and is of same size as that of object.

- d. Object at C i.e. at $2f$. A ray AD from A parallel to principal axis after reflection from mirror passes through F (Rule 1). Another ray AD' from A through F, goes parallel to principal axis i.e. towards D'A' (rule 2) forming real, inverted image of AB at C i.e. at $2f$. The image is of the same size as the object.

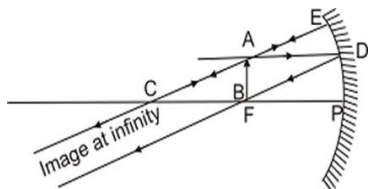


Object at C, Image is also at C. It is real.

- e. Object between F and C (f and 2f) A ray AD from object going parallel to principal axis is reflected towards F (Rule 1). Another ray AE through C is reflected back (Rule 3) forming image of A at A'. Similarly image of B is formed at B'. Image is real, inverted, enlarged and beyond C (2f) i.e. as shown in fig.

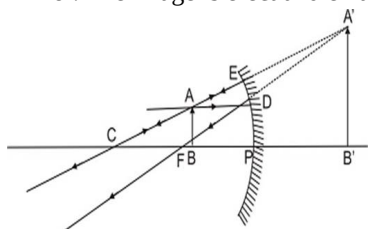


- f. Object at F. A ray AD parallel to principal axis passes through F. Another ray AE strikes the mirror normally at E is reflected back as it passes through C (Rule 3). They form image of object AB at infinity. The image is very much enlarged and is real and inverted.



Object at E. Real, inverted, extremely enlarged image is formed at infinity.

- g. Object between F and P. A ray AD from A goes parallel to principal axis after reflection passes through F (rule 1). Another ray AE striking the mirror normally through C is reflected back (rule 3). They form virtual image of the object behind the mirror. The image is erect and enlarged.



Object between F and P. An erect, enlarged, virtual image is formed behind the mirror.

Images Formed by a Concave mirror

Position of Object	Position of Image	Size of the Image	Nature of Image
At infinity	At focus F	Highly diminished	Real and inverted
At C	At C	Same size	Real and inverted
At F	At infinity	Highly Enlarged	Real and inverted
Beyond C	Between F and C	Diminished	Real and inverted
Between F and C	Beyond C	Enlarged	Real and inverted
Between P and F	Behind the mirror	Enlarged	Virtual and erect

21. i. The distance between the focus and optical centre of lens is called focal length of lens.

ii. Given

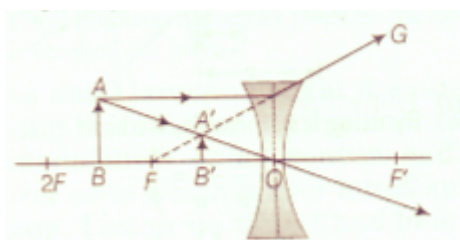
$$f = -30 \text{ cm}, v = -15 \text{ cm}, u = ?, h_o = 5 \text{ cm}$$

$$\frac{1}{f} = \frac{1}{v} - \frac{1}{u} \Rightarrow -\frac{1}{30} = -\frac{1}{15} - \left(\frac{1}{u}\right) \Rightarrow u = -30 \text{ cm}$$

$$\therefore m = \frac{v}{u} = \frac{h_i}{h_o}$$

$$\Rightarrow \frac{15}{30} = \frac{h_i}{5} \Rightarrow h_i = \frac{5}{2} \Rightarrow h_i = 2.5 \text{ cm}$$

iii.



22. a. The four Characteristics of image formed by plane mirror:-

- (i) The image is same size as the object.
- (ii) The image is erect and virtual.
- (iii) The image is laterally inverted.
- (iv) The distance between the object and mirror is same as the distance between image and mirror.

b. Given: height of object $h = 5$ cm, distance of object $u = -20$ cm, focal length of mirror $f = -30$ cm

$$\frac{1}{f} = \frac{1}{v} + \frac{1}{u} \text{ (By using mirror formula)}$$

$$\frac{1}{v} = \frac{1}{-30} - \frac{1}{-20}$$

$$v = 60 \text{ cm}$$

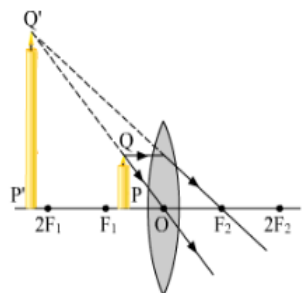
The image is formed at 60 cm on the backside of the mirror.

$$\text{Now, magnification } m = \frac{h'}{h} = -\frac{v}{u}$$

$$h' = 15 \text{ cm}$$

The size of the image is 15 cm

23. i.



ii. U is -ve, V is -ve. By lens formula:

$$\frac{1}{v} - \frac{1}{u} = \frac{1}{f}$$

$$\Rightarrow \frac{-1}{v} - \left(-\frac{1}{u}\right) = \frac{1}{f} \Rightarrow \frac{-1}{v} + \frac{1}{u} = \frac{1}{f}$$

$$\Rightarrow \frac{-u+v}{uv} = \frac{1}{f} \Rightarrow f = \frac{uv}{v-u}$$

This is a required relation between u , v and f in the case when object is placed between optical centre and principal focus of convex lens

iii. Given, $m = -1$

$$u = -20 \text{ cm}$$

$$\therefore m = \frac{v}{u} \Rightarrow -1 = \frac{v}{-20}$$

$$\Rightarrow v = 20 \text{ cm}$$

$$\text{By lens formula, } \frac{1}{v} - \frac{1}{u} = \frac{1}{f}$$

$$\Rightarrow \frac{1}{20} - \left(\frac{-1}{20}\right) = \frac{1}{f} \Rightarrow \frac{1}{20} + \frac{1}{20} = \frac{1}{f}$$

$$\Rightarrow \frac{1}{10} = \frac{1}{f} \Rightarrow f = 10 \text{ cm}$$

$$\therefore \text{Power, } p = \frac{1}{f} = \frac{1}{10 \times 10^{-2}} = 10D$$

$$\Rightarrow P = 10 D$$

24. i. Palmists use a convex lens because it shows an enlarged, virtual and erect image when the object is between F and O of a convex lens.

ii. If the palmist wants a real and magnified image, he should put an object between F_1 and F_2 or on F . But in that case, he will have to use a screen to see the image. So, for convenience, palmists.

iii. Given, $f = 10$ cm, $u = -5$ cm

$$\frac{1}{v} - \frac{1}{u} = \frac{1}{f}$$

$$\text{Or } \frac{1}{v} + \frac{1}{5} = \frac{1}{10}$$

$$\text{Or, } \frac{1}{v} = \frac{1}{10} - \frac{1}{5}$$

$$\text{Or } \frac{1}{v} = \frac{1-2}{10} = -\frac{1}{10}$$

$$\text{Or, } v = -10 \text{ cm}$$

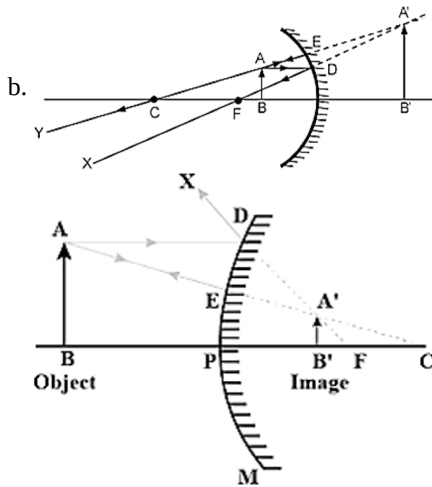
The image is formed at 10 cm on the same side of the lens. It is erect and virtual.

$$\text{Image size} = \frac{v}{u} = \frac{10}{5} = 2$$

Image is twice as big as object.

25. a. Type of mirror

- i. A magnified and virtual image of the object - concave mirror can form virtual magnified image.
- ii. A diminished and virtual image of an object - Convex mirror can form virtual and diminished image.



c. Concave mirrors can also be used to obtain a magnified and real image of an object. When an object is placed anywhere between infinity to focus we get a real and magnified image.

26. Since the image formed is real, hence an inverted image is formed.

size of image, $h_2 = -3$ cm, size of object $h_1 = +2$ cm.

$$\text{Magnification } m = \frac{h_2}{h_1} = \frac{-3}{2} = -1.5. \text{ Also } m = \frac{-v}{u} \text{ or } v = -mu \dots\dots\dots (i)$$

Here $v = -16$ cm (u is always negative)

Substituting in (i), we have $v = -(-1.5)(-16) = -24$ cm. or $v = -24$ cm.

Image is formed 24 cm to the left of the mirror (Negative sign – Image is towards left of mirror)

To calculate focal length. Here $u = -16$ cm, $v = -24$ cm, $f = ?$

$$\text{Using } \frac{1}{f} = \frac{1}{v} + \frac{1}{u} \text{ or } \frac{1}{f} = -\frac{1}{24} - \frac{1}{16} = \frac{-2-3}{48} = -\frac{5}{48} \text{ or } f = -\frac{48}{5} = -9.6 \text{ cm}$$

Negative focal length indicates that the mirror is concave.

27. Object height, $h = +2$ cm

Image height, $h' = -3$ cm (real image hence inverted)

Object distance, $u = -16$ cm

Image distance, $v = ?$

Focal length, $f = ?$

(i) Position of image

From the expression for magnification

$$m = \frac{h'}{h}$$

$$= \frac{v}{u}$$

$$\text{We have, } v = -u \frac{h'}{h}$$

$$\text{Putting values, we get } v = -(-16) \times \frac{-3}{2}$$

$$v = -24 \text{ cm}$$

The image is formed at distance of 24 cm in front of the mirror (negative sign means object and image are on the same side).

(ii) Focal length of mirror

Using mirror formula, Putting values, we get

Using mirror formula,

$$\frac{1}{f} = \frac{1}{u} + \frac{1}{v}$$

$$\text{Putting values, we get } \frac{1}{f} = \frac{1}{-16} + \frac{1}{-24}$$

$$= -\frac{3+2}{48}$$

$$\text{or } f = -\frac{48}{5}$$

$$= -9.6 \text{ cm}$$

28. Given: focal length $f = +20$ cm, object distance $u = -30$ cm, height of object $h_o = 4$ cm.

We know that, $\frac{1}{f} = \frac{1}{v} - \frac{1}{u}$

$$\frac{1}{20} = \frac{1}{v} - \frac{1}{-30}$$

$$\frac{1}{v} = \frac{1}{60}$$

$$\Rightarrow v = 60 \text{ cm}$$

$$\text{So, } \frac{h_i}{h_o} = \frac{v}{u}$$

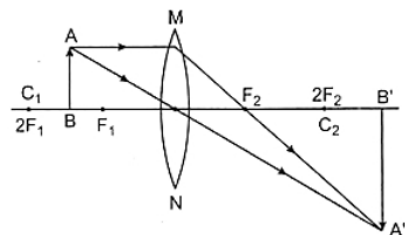
$$\frac{h_i}{h_o} = \frac{60}{-30}$$

$$\Rightarrow -h_i = -8 \text{ cm}$$

Thus, the height or size of the image is 8 cm. The minus sign shows that this height is in the downward direction, that is, the image is formed below the axis.

Ratio of size of image to object = -2

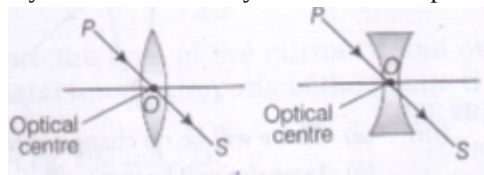
So image is enlarged beyond $2F_2$



Object between F_1 and $2F_1$.

Image is formed beyond $2F_2$, real, inverted.

29. i. The centre point of a lens is known as its optical centre. The optical centre is a point within the lens, directed to which incident rays refract without any deviation in the path whether it is convex lens or concave lens as represented below:



- ii. Given, there is a divergent lens (concave lens.)

Given, $f = -20$ cm, $h_o = 4$ cm, $v = -10$ cm

$\therefore$ By lens formula,

$$\frac{1}{v} - \frac{1}{u} = \frac{1}{f}$$

$$\Rightarrow \frac{-1}{10} - \frac{1}{u} = \frac{-1}{20}$$

$$\Rightarrow \frac{1}{u} = \frac{-1}{10} + \frac{1}{20} \Rightarrow \frac{1}{u} = \frac{-2+1}{20}$$

$$\Rightarrow u = -20 \text{ cm}$$

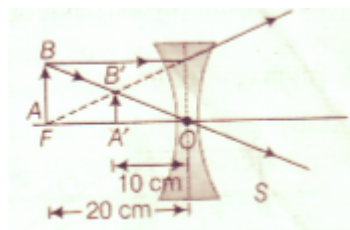
$$\therefore \text{Magnification, } m = \frac{v}{u} = \frac{h_i}{h_o}$$

$$\Rightarrow \frac{h_i}{4} = \frac{-10}{-20}$$

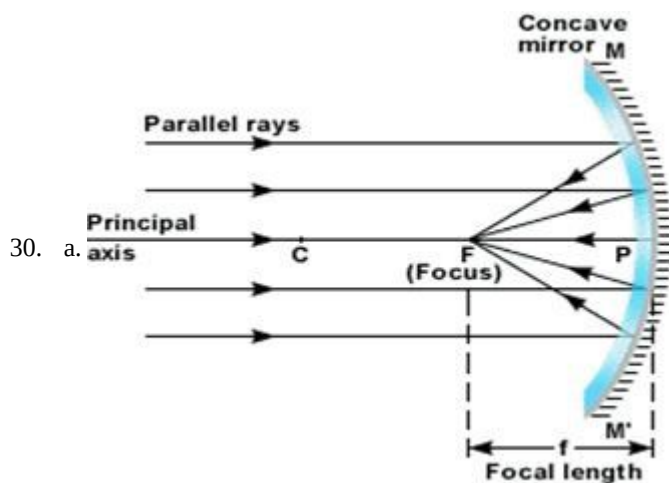
$$\Rightarrow h_i = 2 \text{ cm}$$

Size of the image, $h_i = 2$ cm

- iii.



Thus, the object is placed at 20 cm from the concave lens.



b. Given that $f = -15$ cm ; $u = -25$ cm; $h_o = 4$ cm; $v = ?$

The mirror formula is

$$\frac{1}{f} = \frac{1}{u} + \frac{1}{v}$$

$$\frac{1}{-15} = \frac{1}{-25} + \frac{1}{v}$$

$$\frac{1}{v} = \frac{1}{25} - \frac{1}{15}$$

$$\frac{1}{v} = \frac{3-5}{75} = \frac{-2}{75}$$

$$\Rightarrow v = -37.5 \text{ cm}$$

The screen should be placed 37.5 cm from the pole of mirror and the image is real.

magnification

$$m = \frac{h_i}{h_o} = \frac{-v}{u}$$

$$\frac{h_i}{4} = \frac{-(-37.5)}{-25}$$

$$\Rightarrow h_i = -\frac{37.5}{25} \times 4 = -6 \text{ cm}$$

So the image is enlarged and inverted.

c. Uses of concave mirror:

It is used as a shaving mirror.

It is used as reflector in torch, head light of automobiles etc.

31. Object distance, $u = -15$ cm

Focal length, $f = -10$ cm

Object size, $h = 1$ cm

Image distance, $v = ?$

(i) Position of image

From mirror formula, $\frac{1}{u} + \frac{1}{v} = \frac{1}{f}$

We have, $\frac{1}{v} = \frac{1}{f} - \frac{1}{u}$

$$\text{Putting values, we get } \frac{1}{v} = \frac{1}{-10} - \frac{1}{-15}$$

$$= \frac{-3-(-2)}{30} = -\frac{1}{30}$$

The image is formed at a distance 30 cm on the side of the object. Negative sign indicates that object and image are on the same side.

(ii) Nature of image: The image is in front of the mirror, its nature is real and inverted.

Size of image: From the expression for magnification,

$$m = \frac{h'}{h} = -\frac{v}{u}$$

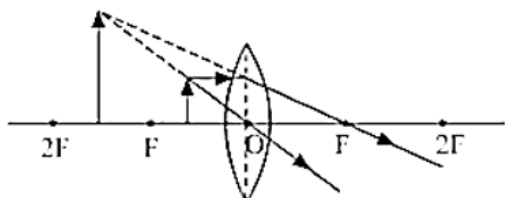
$$\text{We have } h' = -h \times \frac{v}{u}$$

$$\text{putting values, we get } h' = -1 \times \frac{-30}{-15}$$

$$= -2$$

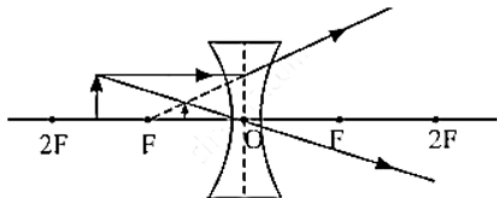
The image formed has size 2 cm and negative sign means inverted and real and enlarged.

32. i. When an object is placed between the optical center and principal focus of a convex lens then image formed beyond $2F$ on the opposite side.



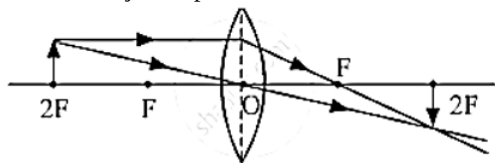
Since the image formed is virtual and erect so sign of magnification will be positive. Moreover, the image formed is magnified therefore the absolute value of magnification will be greater than one.

ii. When an object is placed anywhere in front of a concave lens.



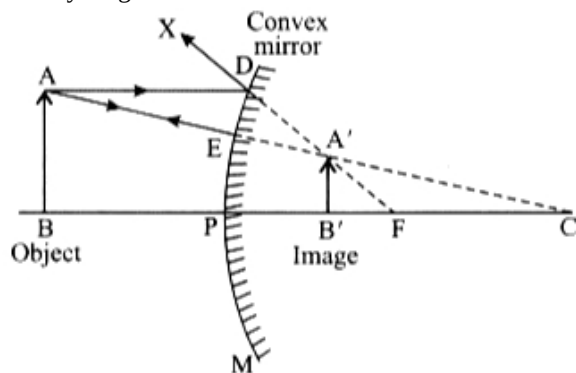
Since the image formed is virtual and erect so sign of magnification will be positive. Moreover, the image formed is diminished therefore the absolute value of magnification will be less than one.

iii. When an object is placed at $2F$ of a convex lens then the image is formed at $2F$ opposite side of the mirror.



33. a. The type of the mirror is convex mirror.

The ray diagram:



Use of convex mirror: A convex mirror always produces a smaller, virtual and erect image of an object. In convex mirror, the length of the image is shorter than that of the object. Hence, it is used as a side view mirror in vehicles.

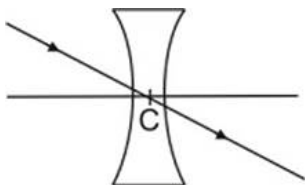
b. **Radius of curvature (R):** The distance between the centre of curvature and pole of a spherical mirror is known as radius of curvature.

Given: Radius of curvature, $R = 24 \text{ cm}$

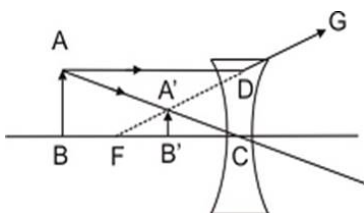
Focal length (f) = $\frac{R}{2} = \frac{24}{2} = +12 \text{ cm}$

Since the focal length is positive, therefore, the nature of the mirror is convex mirror.

34. Rule 1. Incident rays parallel to the principal axis after refraction through concave lens appear to diverge from principal focus F on the same side of the lens.



Rule 2. An incident ray passing through optical centre of a concave lens goes straight after passing through the concave lens.



Formation of images

- a. Object at infinity. In this case, all the rays from infinity will come parallel and appear to diverge from F.
- b. Object anywhere except at Infinity. Let the object AB be at point B. A ray of light from A parallel to the principal axis after refraction diverges towards DG. On producing back, it appears to diverge from the principal focus F. Another ray from A through C, goes undeviated.

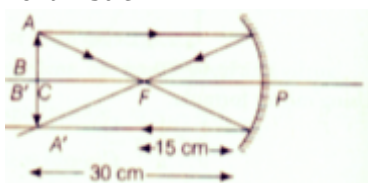
The two rays appear to meet at A'. Thus a virtual, erect, diminished image of AB is formed at A'B' i.e. between C and F.

35. i. When an object is placed at the centre of curvature of concave mirror, the image formed is equal in size to that of object. So, for object distance 20 cm, the mirror C with focal length 10 cm is used as

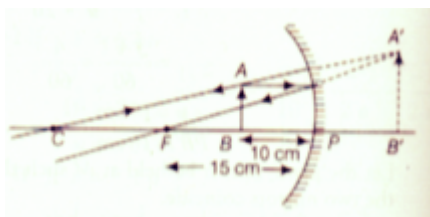
$$u = R = 2f = 2 \times 10\text{cm} = 20\text{cm}.$$

- ii. For shaving purpose, mirror B ($f = 15\text{ cm}$) would be preferred (object distance is 10 cm) Also mirror A ($f=20\text{ cm}$) would be used (object distance is 10 cm) .

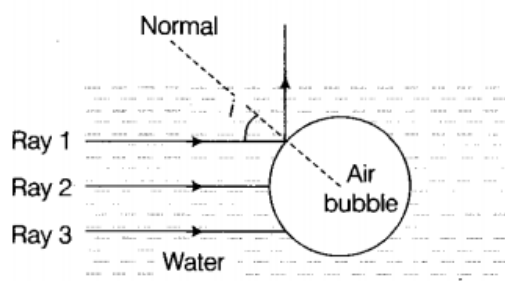
- iii. a. For $u = 30\text{ cm}$



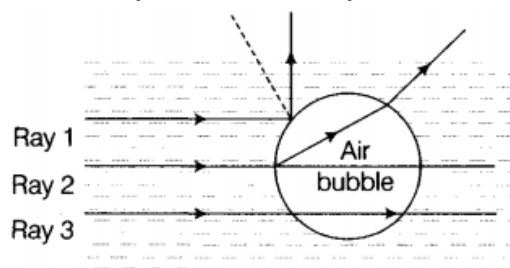
- b. For $u = 10\text{ cm}$



36. i. Diagram for first part is shown as:



- ii. The three rays continue after they meet the air bubble are shown as:



- iii. Refractive index of water is defined as the ratio of the speed of light in a vacuum to the speed of light in water.

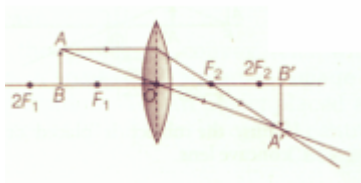
Speed of light in air, $v_a = 3 \times 10^8\text{ ms}^{-1}$

Speed of light in water, $v_w = 2.2 \times 10^8\text{ ms}^{-1}$

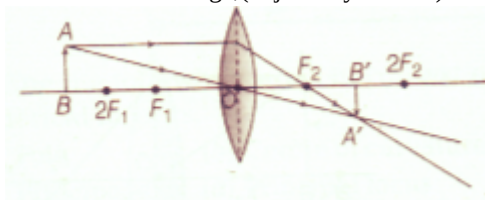
Therefore, Refractive index,

$$n = \frac{v_a}{v_w} = \frac{3 \times 10^8}{2.2 \times 10^8} = 1.4$$

37. i. He should use a convex lens as real images are formed by a convex lens, when objects are placed between the focus and infinity .
- ii. a. For magnified image candle should be placed between focus (F) and centre of curvature (2F) of lens.
b. To get diminished image candle should be placed beyond centre of curvature (2F) of lens.
- iii. a. For magnified image,(object between F and 2F).



- b. For diminished image,(object beyond 2F).



38. i. $u = -60$ cm, $f = -30$ cm, $v = ?$

$$\frac{1}{f} = \frac{1}{v} - \frac{1}{u} \text{ (By using lens formula)}$$

$$\therefore \frac{1}{v} = \frac{1}{f} + \frac{1}{u}$$

$$= \frac{1}{(-30)} + \frac{1}{(-60)} = \frac{-3}{60}$$

$$\therefore v = -20 \text{ cm}$$

- ii. (i) Nature of image: Virtual

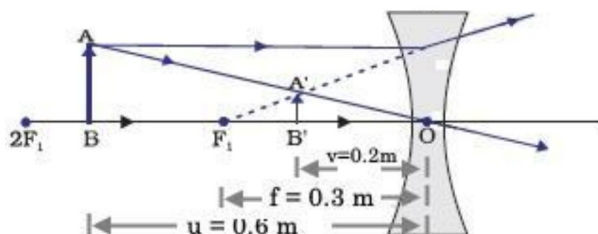
- (ii) Position: 20 cm from lens on the same side as the object.

$$(iii) m = v/u = \frac{-20}{-60} = \frac{1}{3}$$

So, the size of the image is one-third of the object.

- (iv) Image is Erect

- iii. The ray diagram is shown below:



39. Here $\hat{u} = -10$ cm (u is always negative), $v = ?$

$$f = \frac{r}{2} = \frac{+30}{2} \text{ or } f = +15 \text{ cm}$$

f and r for convex mirror are always positive.

Using the formula $\frac{1}{f} = \frac{1}{v} + \frac{1}{u}$ we have $\frac{1}{15} = \frac{1}{v} - \frac{1}{10}$ or

$$\frac{1}{v} = \frac{1}{15} + \frac{1}{10} = \frac{5}{30}v$$

$$v = +6 \text{ cm}$$

or $\therefore$ Image is 6 cm behind the mirror (right or mirror).

It is virtual and erect.

$$\text{Magnification } m = -\frac{v}{u} = \frac{-(+6)}{-10} = 0.6$$

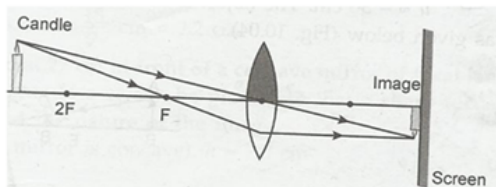
Hence image is 0.6 times the size of the object.

Again $m = \frac{h_2}{h_1}$ where h_2 is the size of the image and h_1 the size of the object.

$$\therefore h_2 = mh_1 = 0.6(5) = 3 \text{ cm.}$$

Hence image is 3 cm. high.

40. When one-half of a convex lens is covered with a black paper, this lens produces a complete image of the object. To prove it we perform experiment:



Take a concave mirror and cover half part of its by using black paper. Place it vertically in a stand. On one side of it place a burning candle. On opposite side of the lens fix a white screen. Adjust the position of candle or screen till clear image of burning candle is formed on the screen. We observe that the image is complete image of the object.

From the experimental observations, we find that image formation does not depend upon the size of a lens. A similar lens can also form complete image of an object placed in front of it. However, brightness of the image decreases when some part of lens is blocked. It is because now lesser number of rays pass through the lens.

41. a. A concave mirror of focal length $f = 10$ cm can produce a magnified real image if an object is placed between F and C (i.e., when $20 \text{ cm} > u > 10 \text{ cm}$) as shown in Fig. (a). However, the same mirror may form a magnified virtual image when object is placed between P and F (i.e., $u < 10 \text{ cm}$) as shown in Fig. (b).



- b. Here focal length of convex mirror $f = +10$ cm, distance of the object from the pole $u = -10$ cm

As per formula,

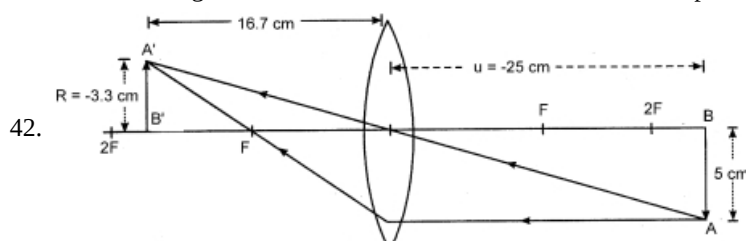
$$\frac{1}{v} + \frac{1}{u} = \frac{1}{f}, \text{ we have}$$

$$\frac{1}{v} = \frac{1}{f} - \frac{1}{u} = \frac{1}{+10} - \frac{1}{-10}$$

$$= \frac{1}{10} + \frac{1}{10} = \frac{1}{5}$$

$$\Rightarrow v = +5 \text{ cm}$$

Thus, image is formed behind the mirror at 5 cm from the pole of mirror.



42.

$h = 5 \text{ cm}$; $h' = ?$, $u = -25 \text{ cm}$ [Object distance is always negative]

$v = ?$; $f = +10 \text{ cm}$ [convex lens]

Using $\frac{1}{f} = \frac{1}{v} - \frac{1}{u}$ or $\frac{1}{v} = \frac{1}{f} + \frac{1}{u}$

$$\frac{1}{v} = \frac{1}{-25} + \frac{1}{10} = \frac{-2 + 5}{50} = \frac{3}{50}$$

$$v = 16.7 \text{ cm}$$

$$m = \frac{h'}{h} = \frac{v}{u} \text{ or } h' = h \frac{v}{u}$$

$$h' = 5 \frac{16.7}{-25} = \frac{-83.5}{5} = -16.7 \text{ cm}$$

$$h' = -3.3 \text{ cm}$$

Negative sign shows image is inverted, real, diminished (3.3 cm) and at 16.7 cm on the right side of lens.

43. Given: Height of object (h_o) = 4 cm

Object distance (u) = -25 cm (-ve as it is in front of mirror)

Focal length (f) = -15 cm

- i. Applying mirror formula and substituting the values,

$$\frac{1}{v} + \frac{1}{u} = \frac{1}{f}$$

$$\frac{1}{v} = \frac{1}{-15} - \frac{1}{-25}$$

$$\frac{1}{v} = \frac{-5+3}{75}$$

$$v = \frac{-75}{2} = -37.5 \text{ cm}$$

The negative sign indicates that the image is in front of the mirror.

Therefore, the screen must be placed in front of the mirror at a distance of 37.5 cm.

ii. Applying the magnification formula and substituting the values,

$$m = \frac{-v}{u} = \frac{h_i}{h_o}$$

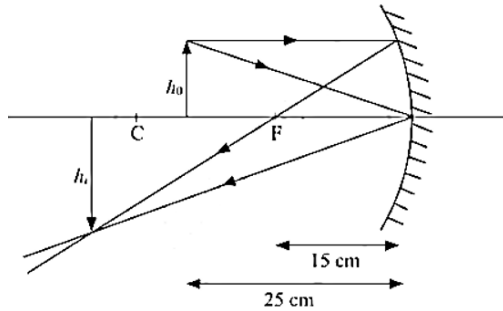
$$\frac{-\left(\frac{-75}{2}\right)}{-25} = \frac{h_i}{4}$$

$$h_i = \frac{-75}{2 \times 25} \times 4$$

$$h_i = -6 \text{ cm}$$

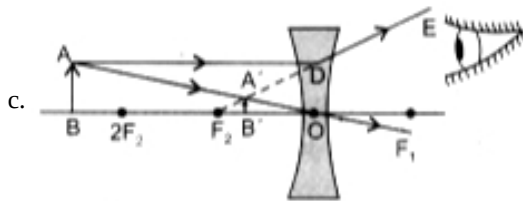
The image will be 6cm high and it will be inverted.

iii. The ray diagram showing the formation of image in this case is,



44. a. The lens L is a concave lens.

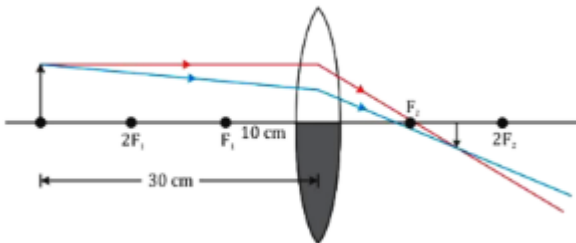
b. The points F_1 and F_2 are called the first focal point and the second focal point respectively.



c. The three characteristics of the image formed by the diagram given above are as follows:

- Virtual and erect,
- diminished
- formed between optical centre O and focus F_2 .

45. When a convex lens is covered half with black paper as shown in diagram, then image of full object will formed , but it will be of less intensity and brightness.



As $h_o = 4 \text{ cm}$, $f = 20 \text{ cm}$ and $u = -15 \text{ cm}$

By lens formula,

$$\frac{1}{f} = \frac{1}{v} - \frac{1}{u}$$

$$\Rightarrow \frac{1}{v} = \frac{1}{f} + \frac{1}{u} = \frac{1}{20} + \frac{1}{(-15)} = \frac{15-20}{300} = \frac{-5}{300}$$

$$\therefore v = -60 \text{ cm}$$

As, magnification,

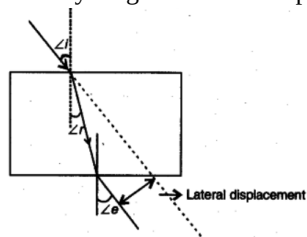
$$m = \frac{h_i}{h_o} = \frac{v}{u}$$

$$\Rightarrow h_i = h_o \times \frac{v}{u} = 4 \times \frac{-60}{-15} = 16 \text{ cm}$$

Image formed is virtual, erect and magnified.

46. i. He should move the screen away from the convex lens to focus the image.

- ii. The size of the image increases.
 - iii. The intensity of the image decreases as the flame moves towards the lens.
 - iv. The flame should be placed at 20 cm from the convex lens on its left side to form inverted image of same size.
47. i. The ray diagram shows the path of a ray of light incident obliquely on one face of a glass slab:



- ii. The glass refractive index is defined as the ratio between the speed of light in the vacuum and the speed of light in the glass.
Refractive index of glass (n_g) = Speed of light in vacuum/speed of light in the glass.

$$n_g = \frac{3 \times 10^8}{2 \times 10^8}$$

$$n_g = 1.5$$

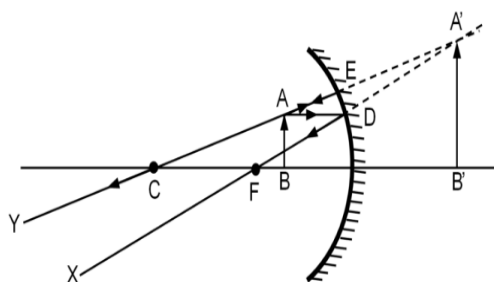
iii. $P = \frac{1}{f(\text{in meter})}$

$$f = \frac{1}{P} = \frac{1}{2.5} = \frac{1}{-25}$$

$$\Rightarrow f = \frac{-1000}{25} \text{ cm} = -40 \text{ cm}$$

The focal length (f) of a concave lens is always negative.

48. i. In concave mirror erect image is formed only when the object is placed between pole and focus. Thus, range to obtain erect image is $0 < u < 12$.
- ii. The image would be larger than the object

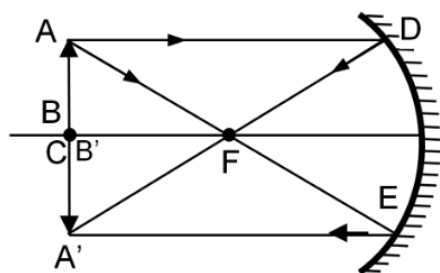


- iii. If the object distance is 24 cm which is the approx position of centre of curvature, then the image will be formed at the same position. Here, $f = -12$ cm, $u = -24$ cm, $v = ?$

By using mirror formula,

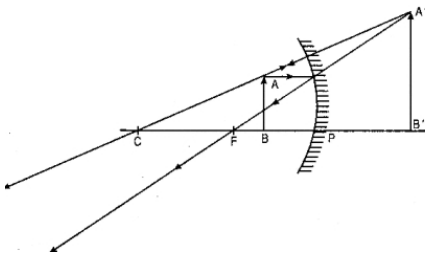
$$\frac{1}{f} = \frac{1}{v} + \frac{1}{u} \Rightarrow \frac{1}{v} = \frac{1}{-12} - \frac{1}{(-24)}$$

$$\Rightarrow v = -24 \text{ cm}$$



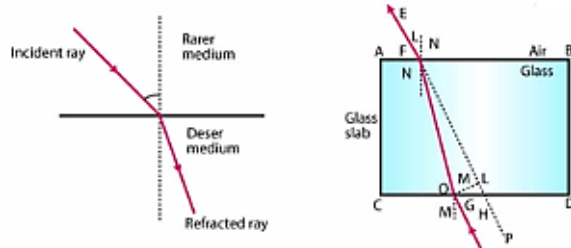
49. When an object is placed in F and F_2 of a convex lens, we get an inverted, enlarged, and the real image is formed beyond $2F_2$ which is on the other side of the lens. Hence we need to place the object between 20 and 40 cm of the lens.
- When an object is placed between F and O of a convex lens, its enlarged, erect and virtual image is formed beyond F_L i.e. on the same side of the lens.
- So for this, we need to place the object at a distance less than 20 cm from the lens.
50. When the object is placed between pole and principal focus of the concave mirror, an erect, enlarged, virtual image is formed behind the mirror. Therefore range of distance is greater than zero and less than focal length i.e. between more than zero to less

than 15 cm.



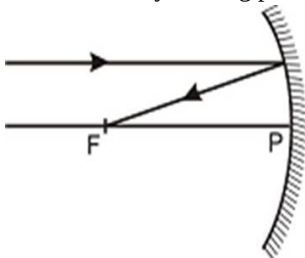
51. Laws of refraction are as follows:

- i. Incident ray, refracted ray and normal at the point of incidence lie in the same plane.
- ii. Ratio of sine of incidence and sine of refraction is constant for the given color and pair of media.

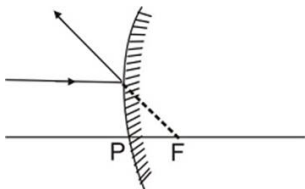


52. There are three rules for formation of images :

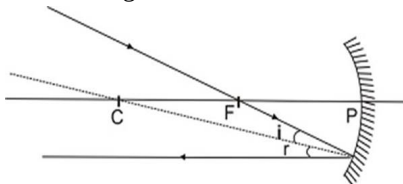
Rule 1. The ray coming parallel to principal axis, after reflection passes through focus F.



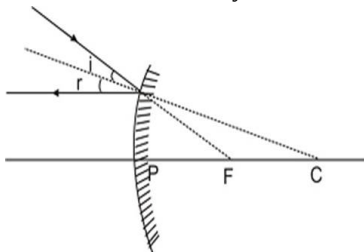
In convex mirror, such rays after reflection appear to diverge from focus since convex mirror is divergent.



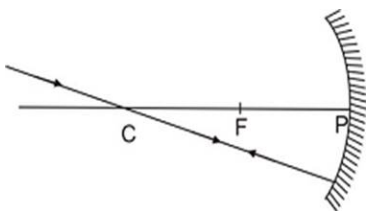
Rule 2 A ray incident on mirror after passing through principal focus F shall become parallel to principal axis after reflection as is shown in fig.



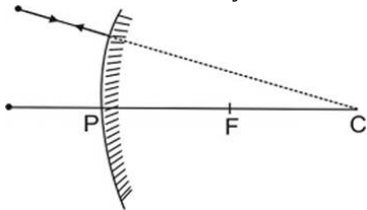
In convex mirror, a ray directed towards F after reflection goes parallel to principal axis as is shown in fig.



Rule 3 A ray incident on mirror after passing through centre of curvature shall be reflected back along the same path as shown in fig.

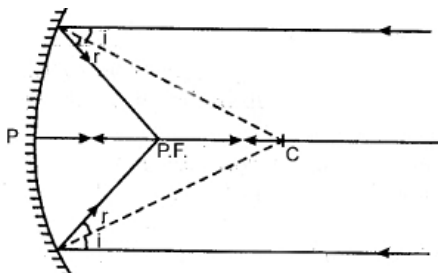


In convex mirror, a ray directed towards, C is reflected back along the same path.

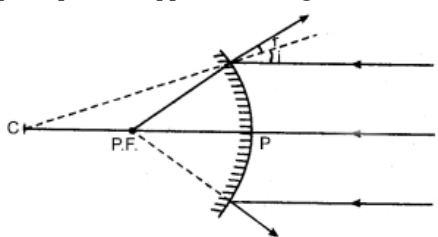


53. Light rays that are parallel to the principal axis of a concave mirror converge at a specific point on its principal axis after reflecting from the mirror. This point is known as the principal focus of the concave mirror.

Principal focus (p.f.) is a point on principal axis of a concave mirror where the rays parallel to principal axis meet after reflection from the mirror.



Additional reading. For convex mirror, principal focus is a point on principal axis of a convex mirror where rays parallel to principal axis appear to diverge from after reflection from the mirror.



54. i. a. Given, $P = +2D$

P is positive, so lens is convex .

$$\therefore f = \frac{1}{P} = \frac{1}{2} = 0.5 \text{ m} = 50\text{cm}$$

b. Given, $P = -4D$

P is negative, so lens is concave .

$$\therefore f = \frac{1}{P} = \frac{1}{-4}$$

$$= -0.25 \text{ m} = -25\text{cm}$$

ii. a. As $u = -100\text{cm}$

Case I $f = 50 \text{ cm}$

Then by lens formula, $-\frac{1}{u} + \frac{1}{v} = \frac{1}{f}$, we get

$$\frac{1}{v} = \frac{1}{f} + \frac{1}{u} = \frac{1}{50} - \frac{1}{100} = \frac{1}{100}$$

$$\text{or } v = 100 \text{ cm}$$

Case II $f = -25 \text{ cm}$

By lens formula, $-\frac{1}{u} + \frac{1}{v} = \frac{1}{f}$,

$$\frac{1}{v} = \frac{1}{f} + \frac{1}{u} = -\frac{1}{25} - \frac{1}{100} = \frac{-5}{100} = -\frac{1}{20}$$

$$v = -20 \text{ cm}$$

b. As, $m = \frac{v}{u}$

Case I: $u = -100 \text{ cm}$, $v = 100 \text{ cm}$

$$m = \frac{100\text{cm}}{-100\text{cm}} = -1$$

Case II: $u = -100 \text{ cm}$, $v = -20 \text{ cm}$

$$m = \frac{-20\text{cm}}{-100\text{cm}} = \frac{1}{5} = 0.2$$

55.	Concave Mirror	Convex Mirror
	A concave mirror is made by silvering the outer surface of a part of a hollow sphere and reflection takes place from the inner surface.	A convex mirror is made by silvering the inner surface of a part of a hollow sphere and reflection takes place from the outer bulging surface.
	Concave mirrors are converging mirrors. The light rays incident on a concave mirror converges after reflection.	A convex mirror is a diverging mirror. The light rays incident on a convex mirror diverges after reflection.
	The image formed by it is real as well as virtual depending on the position of the object from the mirror.	The image formed by it is always virtual for all positions of the object in front of the mirror.

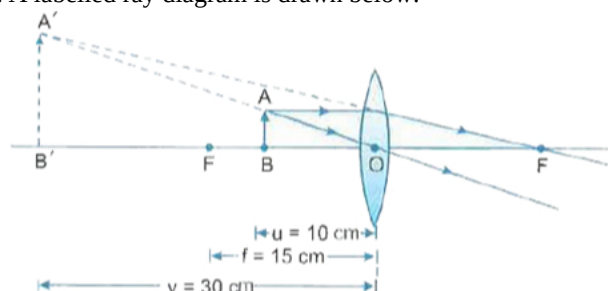
56. i. a. Power of a lens is a measure of the degree of convergence or divergence of light rays achieved by it. Mathematically, power (P) of a lens is defined as reciprocal of its focal length i.e., $P = \frac{1}{F}$.

b. Principal focus of a concave mirror is a point situated on its principal axis, where light rays coming parallel to the principal axis of mirror, after reflection, actually converge.

ii. a. $\frac{1}{v} - \frac{1}{u} = \frac{1}{F}$

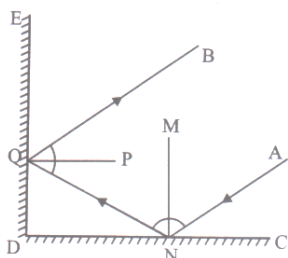
b. $\frac{1}{v} + \frac{1}{u} = \frac{1}{F}$

iii. A labelled ray diagram is drawn below:



A virtual, erect and enlarged image is being formed on the same side of the lens at a distance of 30 cm from optical centre of lens.

57. When two plane mirrors are at 90° then incident ray and reflected ray will always be parallel as explained below:



ED and DC are two plane mirrors placed at 90° :

$$\angle ANM = \angle MNQ \text{ (angle } i = \text{angle } r)$$

$$\angle NQP = \angle PQB \text{ (same as above)}$$

$$\angle MNQ + \angle QND = 90^\circ \text{ (MN is normal on CD) ... (1)}$$

$$\angle QDN = 90^\circ \text{ (mirrors are at right angle)}$$

Hence, in $\triangle QDN$;

$$\angle QND + \angle NQD = 90^\circ$$

(acute angles of a right triangle are complementary) ... (2)

From equations (1) and (2);

$$\angle MNQ = \angle NQD$$

$$\text{Hence, } \angle QND = \angle NQP$$

(because $\angle NQP$ and $\angle NQD$ are complementary)

Now, $\angle PQB$ and $\angle PQD$ are supplementary (PQ is normal on DE)

$$\text{So, } \angle BQE = \angle QND = \angle MNQ = \angle ANM$$

$$\text{Or, } \angle BQE = \angle ANM$$

Since corresponding angles are equal

Hence, $BQ \parallel AN$ proved

This means that incident ray and reflected ray will always be parallel; irrespective of value of angle of incidence.

58. Magnification of a spherical mirror is the ratio of size of the image formed by the size of the object.

$$\text{Magnifying power } m = \frac{\text{size of the image}(h_2)}{\text{size of the object}(h_1)}$$

Since $\angle i = \angle r$ So, ABP and A'B'P' are similar

$$\therefore \frac{A'B'}{AB} = \frac{PB'}{PB} \dots\dots(i)$$

Now AB = + h₁ (Measured upward)

A'B' = -h₂ (Measured downward)

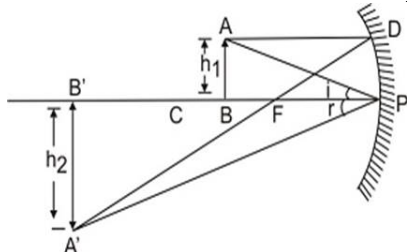
PB = - u (Measured against incident light)

PB' = - v

Substituting these values in equation (i) we have

$$-\frac{h_2}{h_1} = \frac{-v}{-u} \text{ or } m = \frac{h_2}{h_1} = \frac{-v}{u}$$

$$\text{Similarly for convex mirror } m = \frac{h_2}{h_1} = \frac{-v}{u}$$



Magnification is positive when image is erect and virtual. Magnification, m is negative when image is inverted and real.

Magnification of convex, as well as plane mirror, is positive. Magnification of plane mirror is + 1 since the image is erect.

59. Height of the object h = + 2.0 cm; Focal length f = + 10 cm; object-distance u = -15 cm; Image-distance v = ? Height of the image h' = ?

$$\text{Since } \frac{1}{v} - \frac{1}{u} = \frac{1}{f}$$

$$\text{or, } \frac{1}{v} = \frac{1}{u} + \frac{1}{f}$$

$$\frac{1}{v} = \frac{1}{(-15)} + \frac{1}{10} = -\frac{1}{15} + \frac{1}{10}$$

$$\frac{1}{v} = \frac{-2+3}{30} = \frac{1}{30}$$

$$\text{or, } v = + 30 \text{ cm}$$

The positive sign of v shows that the image is formed at a distance of 30 cm on the other side of the optical centre. The image is real and inverted.

$$\text{Magnification } m = \frac{h'}{h} = \frac{v}{u} \text{ or, } h' = h \frac{v}{u}$$

$$\text{Height of the image, } h' = (2.0)\left(\frac{+30}{-15}\right) = -4.0 \text{ cm}$$

$$\text{or } m = \frac{+30\text{cm}}{-15\text{cm}} = -2$$

The negative signs of m and h' show that the image is inverted and real. It is formed below the principal axis. Thus, a real, inverted image, 4 cm tall, is formed at a distance of 30 cm on the other side of the lens. The image is two times enlarged.

60. Position of the candle flame = 12.0 cm

Position of the lens = 50.0 cm

Position of the screen = 88.0 cm

$$\text{i. } u = 50 - 12 = 38 \text{ cm}$$

$$\text{Image distance } v = 88 - 50 = 38 \text{ cm}$$

$$\text{Focal length} = \frac{1}{v} - \frac{1}{u} = \frac{1}{t}$$

$$f = 19 \text{ cm}$$

$$\text{ii. Object distance } u = 50 - 31 = 19 \text{ cm}$$

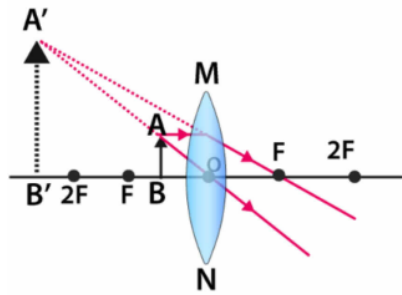
Here

Object distance = focal length

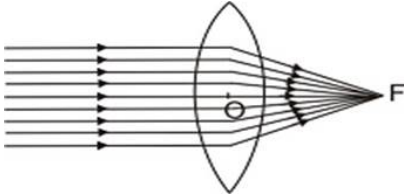
Hence the image is formed at infinity.

iii. If he further shifts the candle towards the lens. The object comes between F and O. In this case. The image is virtual, enlarged and erect and is formed on the same side of the lens.

iv.

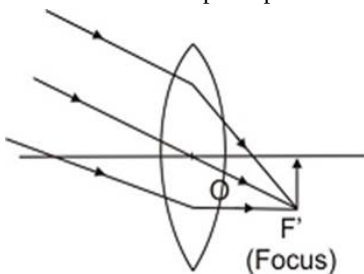


61. Object at Infinity. When object is at infinity, a real image is formed at F on the other side of the lens

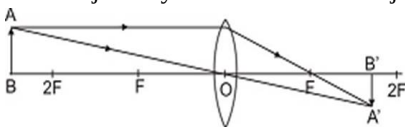


Object at infinity. Image at F on the other side of lens.

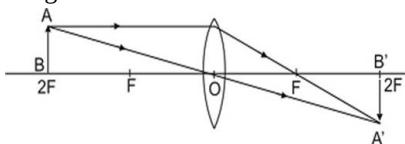
However if the rays are parallel to themselves but not parallel to principal axis, then these rays after refraction will form image at focus F' and not at principal focus F.



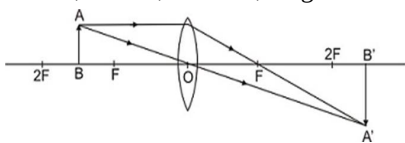
Object at infinity, rays parallel to themselves but not parallel to principal axis. Image is formed at F' , the focus on the other side of lens. Object beyond $2F$. When the object is beyond $2F$, a real, inverted, diminished image is formed between F and $2F$.



Object beyond $2F$, real, inverted, diminished image between F and $2F$. Object at $2F$. When the object is at $2F$, a real, inverted image of the same size is formed on the other side of the lens at $2F$ as given in Fig.

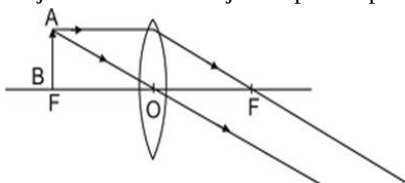


Object at $2F$, image at $2F$ on the other side of the lens. Image is of size same as that of the object. When the object is between F and $2F$, its real, inverted, magnified image is formed on the other side of the convex lens as shown in fig.



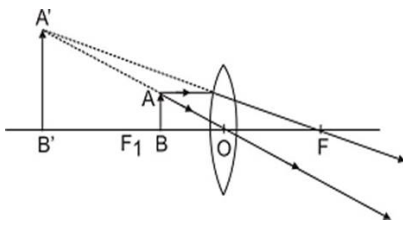
Object between F and $2F$ real, inverted, magnified image is formed beyond $2F$ on the other side of lens.

Object at F. When object is placed principal focus, a real, inverted, very highly magnified image is formed at infinity.



Object at F, a very highly magnified, real, inverted image is formed at infinity.

Object between F and C. When an object is placed between principal focus and optical centre of the lens, virtual, erect, magnified image is formed on the same side of the lens.

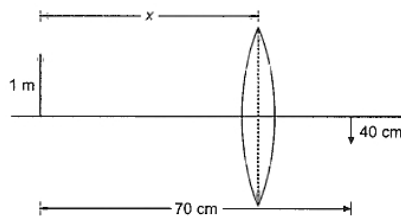


Object between F and C; a virtual, erect, magnified image is formed on the same side.

62. Since, the image is formed on the screen, so the image is real and inverted.

Given: $h = 1\text{ m} = 100\text{ cm}$, height of image $h' = -40\text{ cm}$

Let the object be kept at a distance x from the lens which height is 1 m



$$\therefore v = +(70 - x)$$

$$\text{Now } m \frac{h'}{h} = \frac{v}{u}$$

$$\therefore \frac{-40}{+100} = \frac{(70-x)}{-x}$$

$$\text{or } 40x = 7000 - 100x$$

$$\text{i.e., } x = 50\text{ cm}$$

$$\therefore u = -x = -50\text{ cm}$$

$$\text{and } v = 70 - x = 70 - 50 = 20\text{ cm}$$

Substituting the values of u and v in the lens formula,

$$\frac{1}{v} - \frac{1}{u} = \frac{1}{f}$$

$$\text{We have, } \frac{1}{20} - \frac{1}{-50} = \frac{1}{f}$$

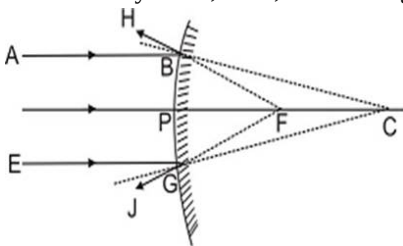
$$\therefore f = +\frac{110}{7}$$

$$= +14.3\text{ cm}$$

The Focal length of the lens = 14.3 cm .

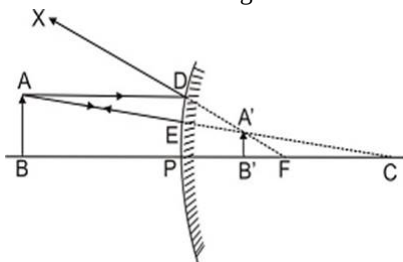
63. Object at Infinity: Rays from infinity come parallel to principal axis. Ray AB is reflected towards BH and ray EG is reflected towards GJ. When produced back they meet at F, the principal focus (Rule 1).

An extremely small, erect, virtual image is formed at F.

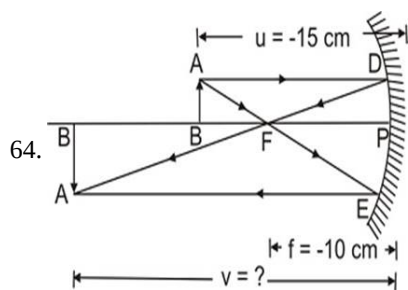


Object at infinity: A virtual, erect, extremely diminished image is formed at F.

Object placed anywhere except infinity. A ray AD parallel to principal axis after reflection appears to pass through F. Another ray AE normal to mirror and appearing to pass through C is reflected back along the same path. They appear to meet at A' forming image of A and A'. Similarly, image of B is formed at B'. This virtual, erect image of AB is formed at A'B' between P and F behind the mirror. Image is erect and diminished in size.



Object any where, A virtual, erect, diminished image is formed behind the mirror between F and P



Here $u = -15 \text{ cm}$ (Object is to the left of P)

$f = -10 \text{ cm}$ (F is to the left of P)

$v = ?$

size of object $= h_1$

size of image $h_2 = ?$

Using $\frac{1}{f} = \frac{1}{v} + \frac{1}{u}$

$$-\frac{1}{10} = \frac{1}{v} - \frac{1}{15}$$

$$\frac{1}{v} = \frac{1}{15} - \frac{1}{10} = \frac{2-3}{30} = -\frac{1}{30} \text{ or } v = -30 \text{ cm}$$

Position of image is 30 cm to the left of P since negative sign of v indicates that the image is on the same side of the mirror.

$$\text{Magnification } m = \frac{h_2}{h_1} = -\frac{v}{u} \text{ or } m = -\frac{(-30)}{(-15)} = -2$$

Negative magnification indicates that the image is inverted, two times magnified and real.

$$\text{Again } m = \frac{h_2}{h_1} = -2 \text{ or } h_2 = -2h_1$$

where h_2 is the size of the image and h_1 , the size of the object.

Here $h_1 = 1 \text{ cm}$. $h_2 = (-2) 1 = -2 \text{ cm}$.

Size of the image is 2 cm and is inverted.

65. The modern Cartesian sign conventions are :

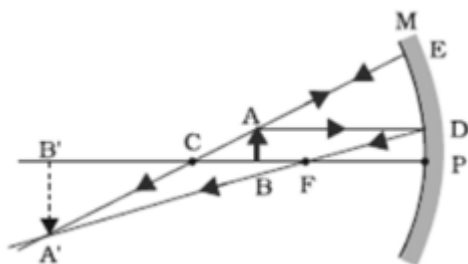
- All the rays are drawn from left to right.
- All distances are measured from the pole of the spherical mirror.
- The distances measured in the direction of the incident light are taken as positive and those measured in a direction opposite to the direction of incident light are taken as negative.
- The heights measured upwards and perpendicular to the principal axis of the mirror are taken as positive and vice versa.

Since the object is always placed on the left of the mirror, u is always negative.

If u is the distance of object and v the distance of image from pole of the mirror, then

$$\frac{2}{R} = \frac{1}{v} + \frac{1}{u} \text{ or } \frac{1}{f} = \frac{1}{v} + \frac{1}{u}$$

66. i. The student should use a Concave mirror because a concave mirror produces real images.
 ii. To get magnified image, the student should put the candle flame between f and $2f$.
 iii. The ray diagram will be as follows:



- iv. Yes, concave mirror can be used to obtain a diminished image. When the object is placed beyond $2f$, then the image formed will be diminished one.

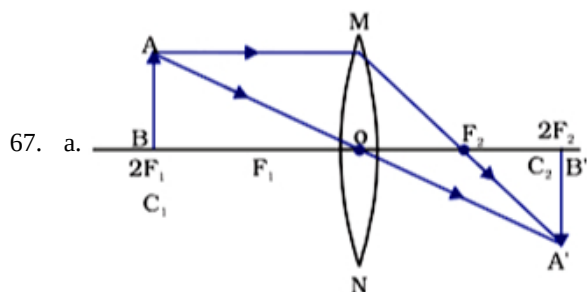


Image will be of same size as the object, real and inverted.

- b. i. Applying mirror formula.

$$\frac{1}{f} = \frac{1}{v} + \frac{1}{u}$$

$$\therefore \frac{1}{v} = \frac{1}{f} - \frac{1}{u}$$

$$= \frac{1}{(-12)} - \frac{1}{(-18)} = \frac{-1}{12} + \frac{1}{18} = \frac{-3+2}{36} = \frac{-1}{36}$$

$$v = -36 \text{ cm}$$

$$\text{ii. } h' = -\frac{v}{u} \times h = -\frac{-36 \text{ cm}}{-18 \text{ cm}} \times 1.5 \text{ cm}$$

$$= -3 \text{ cm (Magnified, inverted image)}$$

68. a. A transparent material bounded by two surfaces of which one or both surfaces are spherical / curved.

a. Converging lens

b. Diverging lens

Double concave lens is a diverging lens.

- b. To find the position of the object, we can use the lens formula:

$$f = 15 \text{ cm, } v = 20 \text{ cm, } u = ?$$

$$\frac{1}{f} = \frac{1}{v} - \frac{1}{u}$$

$$\Rightarrow \frac{1}{u} = \frac{1}{v} - \frac{1}{f}$$

$$= \frac{1}{20 \text{ cm}} - \frac{1}{15 \text{ cm}}$$

$$= \frac{4-3}{60}$$

$$u = -60 \text{ cm.}$$

image is diminished.

69. If u is the distance of object and v , the distance of image from optical centre of the lens, then focal length f is related to u and v by $\frac{1}{f} = \frac{1}{v} - \frac{1}{u}$ which is called lens formula. The formula is equally applicable to convex, concave or any other type of lens. (e.g. concavo-convex, plano-convex, convexo-concave, plano-concave etc.)

Sign Conventions:

(1) The object is always placed to the left of the lens.

(2) All distances parallel to principal axis are measured from optical centre of the lens.

(3) All distance measured to the right of optical centre (along x-axis) are taken as positive while those measured to the left of optical centre (along x-axis) are taken as negative.

(4) Distance measured perpendicular to and above the principal axis (along y axis) are taken as positive.

(5) Distance measure perpendicular to and below the principal axis (y'-axis) are taken as negative.

Assumptions (1) Object is taken on principal axis. (2) The lens is thin.

70. i. Given:

distance of image from the lens, $i = 10 \text{ cm}$

power of the lens, $P = -25 \text{ D}$

Now the focus of the lens:

$$P = \frac{1}{f}$$

where:

f = focal length

$$-25 = \frac{1}{f}$$

$$f = -0.04 \text{ m} = -4 \text{ cm}$$

From the equation of lens:

$$\frac{1}{f} = \frac{1}{i} + \frac{1}{o}$$

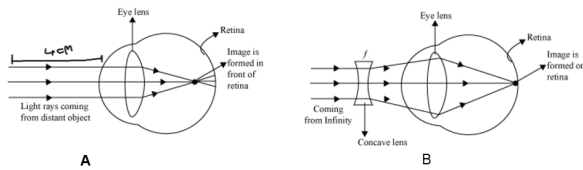
where:

o = distance of the object

$$-\frac{1}{4} = \frac{1}{10} + \frac{1}{o}$$

$\rho = -\frac{20}{7}$ cm i.e. negative sign means that the image formed is on the same side as that of the object.

ii.



71. The ability of a lens to converge or diverge light rays is called power of the lens. It is defined as the reciprocal of focal length. It's SI unit is dioptre (D). If focal length is expressed in metres, then power is expressed in dioptre. We can say, dioptre is the power of a lens whose focal length is one metre. For concave lens P and f are negative. For convex lens P and f are positive. Lens A of focal length + 10 cm is convex lens

$$\text{and power, } P = \frac{100}{f(\text{in cm})} = \frac{100}{10} = +10D$$

Lens B of focal length - 10 cm is concave lens

$$\text{and power, } P = \frac{100}{f(\text{in cm})} = \frac{100}{-10} = -10D$$

Lens A (i.e. convex lens) will form a virtual and magnified image of an object placed 8 cm from it, as shown.

